

12.268

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

A 3×3 matrix has elements such that its trace is 11 and its determinant is 36. The eigenvalues of the matrix are all known to be positive integers. Find the largest eigenvalue of the matrix.

Solution:

Let the eigenvalues of the 3×3 matrix $\mathbf{A}$ be $\lambda_1, \lambda_2, \lambda_3$.

We are given:

$$\text{trace}(\mathbf{A}) = \lambda_1 + \lambda_2 + \lambda_3 = 11, \quad (1)$$

$$\det(\mathbf{A}) = \lambda_1 \lambda_2 \lambda_3 = 36 \quad (2)$$

$$\lambda_1, \lambda_2, \lambda_3 > 0 \quad (3)$$

Check the possible factor triplets of 36 and their sums:

$$1 + 4 + 9 = 14 \neq 11 \quad (4)$$

$$2 + 2 + 9 = 13 \neq 11 \quad (5)$$

$$2 + 3 + 6 = 11 \quad (6)$$

$$3 + 3 + 4 = 10 \neq 11 \quad (7)$$

Hence, the eigenvalues are:

$$\begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}. \quad (8)$$

Thus, the largest eigenvalue is:

$$\lambda_{\max} = 6.$$